Zeitschrift für Physik 206, 205—211 (1967)

An Empirical Formula for the Electron-Impact Ionization Cross-Section*

WOLFGANG LOTZ
Institut für Plasmaphysik, Garching bei München

Received May 24, 1967

An empirical formula with three free parameters is given for the representation of cross-sections for single ionization of atoms and ions from the ground state by electron-impact. Almost all experimental results can be approximated by this formula within 10% over the whole energy range between threshold and 10 keV; all experimental results can be approximated within experimental error. The formula proposed is not suitable to reproduce the exact contour of fine structures in the ionization cross-section curve.

1. Introduction

In the course of time numerous authors have attempted to represent the functional dependence of the electron-impact ionization cross-section by a theoretical or empirical formula. As examples I should like to mention THOMSON[1], BETHE[2], ELWERT[3], KNORR[4], GRYZINSKI[5], and DRAWIN[6,7]; but these attempts shall not be discussed here further as DRAWIN has already done so excellently. As a summary it may be stated that up to now no formula is known with which it is possible to represent all experimental data within experimental error. Especially for the ionization cross-section of neon (Fig. 3), discrepancies are found that exceed a factor of two. The best approximation has been achieved by DRAWIN.

In 1965 RUDGE and SCHWARTZ[8,9] presented theoretical calculations for He^+: Theory and experiment gave identical results within an experimental error of about 10% (Fig. 2), an accomplishment that renders the

* This work was performed as part of the joint research program of the Institut für Plasmaphysik, Garching, and EURATOM.

[1] THOMSON, J. J.: Phil. Mag. **23**, 449 (1912).
[2] BETHE, H. A.: Ann. Physik **5**, 325 (1930).
[3] ELWERT, G.: Z. Naturforsch. **7**a, 432 (1952).
[4] KNORR, G.: Z. Naturforsch. **13**a, 941 (1958).
[5] GRYZINSKI, M.: Phys. Rev. **115**, 374 (1959).
[6] DRAWIN, H. W.: Z. Physik **164**, 513 (1961).
[7] DRAWIN, H. W.: Z. Physik **172**, 429 (1963).
[8] RUDGE, M. R. H., and S. B. SCHWARTZ: Proc. Phys. Soc. (London) **86**, 773 (1965).
[9] RUDGE, M. R. H., and S. B. SCHWARTZ: Proc. Phys. Soc. (London) **88**, 563 (1966).

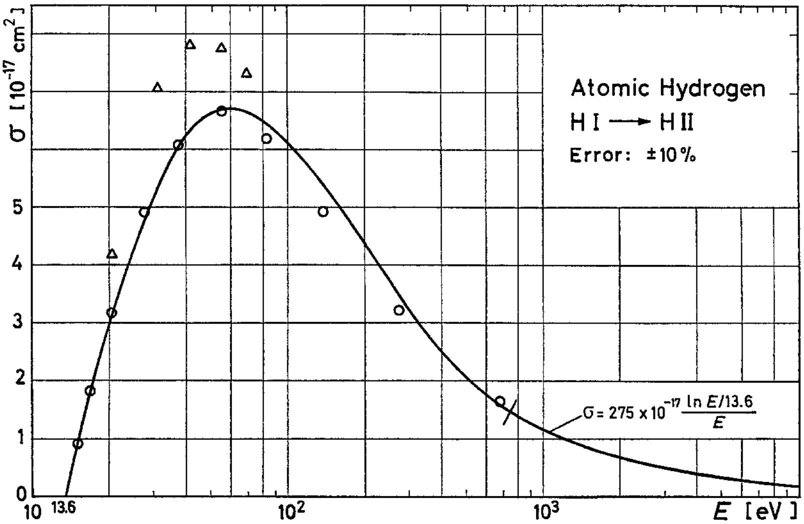

Fig. 1. The curve gives mean experimental values of the electron-impact ionization cross-section from the ground state of atomic hydrogen (Lotz[10]). Values calculated by applying the constants of Table 1 are marked by small circles. These values have purposely not been interconnected by a curve, because the experimental curve and the calculated curve nearly coincide. Values calculated by Rudge and Schwartz[9] are marked by triangles. The experimental error given is a typical value near maximum cross-section (about two times probable error)

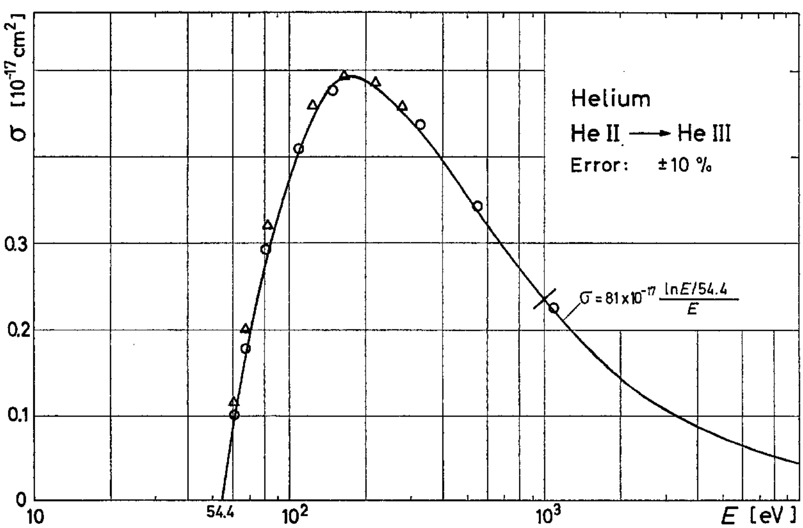

Fig. 2. The curve gives mean experimental values of the electron-impact ionization cross-section from the ground state of singly ionized helium (Lotz[10]). For explanations see Fig. 1!

[10] Lotz, W.: Astrophys. J., Suppl. **14**, 207 (1967).

results given by RUDGE and SCHWARTZ for a hypothetical ion with $Z=128$ rather trustworthy; the reduced cross-sections for He^+ and for the ion $Z=128$ are not much different. For hydrogen there still is a discrepancy beyond experimental error, but this discrepancy is not large (Fig. 1).

2. The New Formula

All electrons of an atom or an ion contribute with individual cross-sections σ_{ei} to the total ionization cross-section σ as far as the kinetic energy E of the impacting electron is greater than the binding energy P_i of the bound electrons. All q_i equivalent electrons of a subshell contribute equally to the subshell cross-section σ_i:

$$\sigma=\sum_{i=1}^{N}\sigma_i; \qquad \sigma_i=q_i\sigma_{ei}; \tag{1}$$

where N is the number of subshells present.

The results of the calculations by BETHE[2] and RUDGE and SCHWARTZ[9] suggest the following ansatz for single ionization from the ground state:

$$\sigma_i=a_i\,q_i\,\frac{\ln(E/P_i)}{E\,P_i}, \qquad E\geqq P_i. \tag{2}$$

If $E\gg P_i$, then a_i is a constant between 2.6 and $4.5\times10^{-14}\,\text{cm}^2\,(\text{eV})^2$ experimentally[10].

For electron energies E smaller than the value E_m, for which the cross-section reaches a maximum, a_i is a certain function of E. Extensive trials with different approximations eventually yielded the following still relatively simple formula with three free parameters ($E\geqq P_i$):

$$\sigma_i=a_i\,q_i\,\frac{\ln(E/P_i)}{E\,P_i}\,\{1-b_i\exp[-c_i(E/P_i-1)]\}. \tag{3}$$

Here a_i, b_i, and c_i are individual constants. The total cross-section for single ionization from the ground state can then be written as:

$$\sigma=\sum_{i=1}^{N}a_i\,q_i\,\frac{\ln(E/P_i)}{E\,P_i}\,\{1-b_i\exp[-c_i(E/P_i-1)]\}; \qquad E\geqq P_i. \tag{4}$$

$i=1$ is the outermost shell, $i=2$ is the next inner subshell, etc.; P_1 is the ionization potential.

Near threshold $E\approx P_1$, Formula (4) reduces to

$$\sigma\approx a_1\,q_1\,\frac{(E/P_1-1)}{P_1^2}\,(1-b_1)\propto U-1, \tag{5}$$

with $U=E/P_1$. For large electron energies $E \gg P_i$, Formula (3) becomes

$$\sigma_i = a_i q_i \frac{\ln(E/P_i)}{E P_i} \propto \frac{\ln E}{E}. \tag{6}$$

Thus Formulae (3) and (4) give the correct energy dependence both for small and for large energies of the impacting electron.

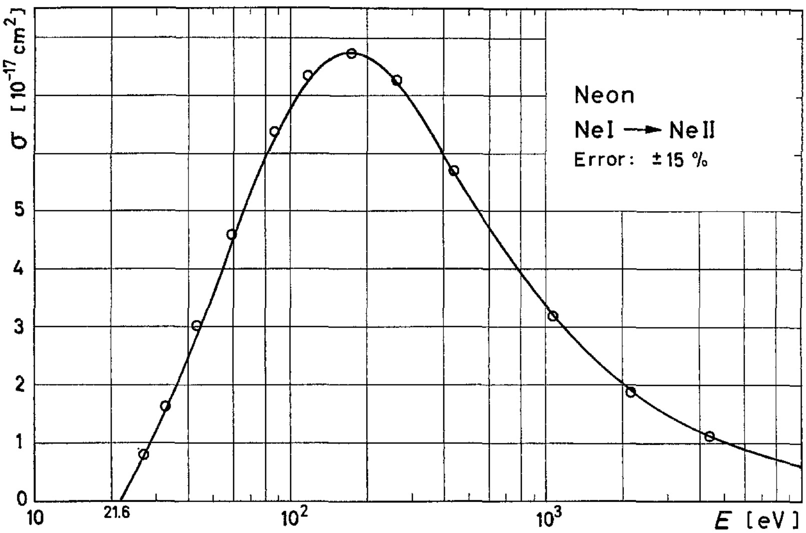

Fig. 3. The curve gives mean experimental values of the electron-impact ionization cross-section for single ionization from the ground state of neon (Lotz[10]). Values calculated by applying the constants of Table 1 are marked by small circles. The experimental error given is a typical value near maximum cross-section (about two times probable error)

For practical purposes Formula (4) can in most cases be substituted by a simpler formula:

$$\sigma = a\{1 - b \exp[-c(U-1)]\} \sum_{i=1}^{N} q_i \frac{\ln(E/P_i)}{E P_i}; \qquad E \geqq P_i. \tag{7}$$

For most atoms and ions $N=2$ gives satisfactory results; for krypton and xenon, $N=3$ gives a better approximation for large energies E, because here the ten d electrons of the second next inner subshell contribute appreciably. For sodium and mercury, Formula (7) cannot be used, because the $2p$ electrons or the $5d$ electrons respectively of the next inner subshell deviate to a large extent from Formula (2). In these cases Formula (4) gives better results.

3. Comparison with Experiment

For all experimental curves collected by the author[10], the constants *a*, *b*, and *c* have been determined for best fit and can be found in Table 1. Binding energies given in columns 3, 4, and 5 of Table 1 have been taken from recent papers of the author[11,12].

Table 1. *Relevant data for all experimentally well known electron-impact ionization cross-sections for single ionization from the ground state. Constants a, b, and c are used in formula (7) [sodium and mercury: formula (4)] to calculate cross-sections*[13], *which are marked in Fig. 1 through 4 by small circles. Constant a is given in* 10^{-14} cm^2 $(eV)^2$

Species	Configuration	Binding energies in eV			Constants		
		P_1	P_2	P_3	a	b	c
H I	$1s$	13.6	—	—	4.0	0.60	0.56
He I	$1s^2$	24.6	—	—	4.0	0.75	0.46
He II	$1s$	54.4	—	—	4.4	0.38	0.60
Li I	$1s^2\,2s$	5.39	58	—	4.0	0.70	2.4
Li II	$1s^2$	75.6	—	—	4.0	0.48	0.60
N I	$2s^2\,2p^3$	14.5	20.3	—	3.4	0.83	0.22
N II	$2s^2\,2p^2$	29.6	36.7	—	4.1	0.46	0.62
O I	$2s^2\,2p^4$	13.6	28.5	—	2.8	0.74	0.24
Ne I	$2s^2\,2p^6$	21.6	48.5	—	2.8	0.92	0.19
Ne II	$2s^2\,2p^5$	41.1	66.4	—	3.2	0.83	0.48
Na I	$3s$	5.14	—	—	4.0	0	—
	$2p^6$	—	34	—	3.6	0.84	0.32
Na II	$2s^2\,2p^6$	47.3	80.1	—	3.6	0.84	0.32
A I	$3s^2\,3p^6$	15.8	29.2	—	4.0	0.62	0.40
K I	$3p^6\,4s$	4.34	18.7	—	4.0	0	—
K II	$3s^2\,3p^6$	31.7	47.9	—	3.8	0	—
Kr I	$3d^{10}\,4s^2\,4p^6$	14.0	27.5	93	4.0	0.71	0.76
Rb I	$4p^6\,5s$	4.18	15.3	—	4.0	0	—
Xe I	$4d^{10}\,5s^2\,5p^6$	12.1	23.4	67	4.0	0.54	0.64
Cs I	$5p^6\,6s$	3.89	12.3	—	4.0	0	—
Hg I	$6s^2$	10.4	—	—	4.0	0.55	4
	$5d^{10}$	—	14.8	—	4.0	0.88	0.16

The experimental curves of Figs. 1–4 are identical with those given in References [10] and [13] with the exception of argon. The recent paper of

[11] Lotz, W.: J. Opt. Soc. Am. **57**, 873 (1967).

[12] Lotz, W.: Report IPP 1/56, Institut für Plasmaphysik, Garching bei München 1967.

[13] Lotz, W.: Report IPP 1/50, Institut für Plasmaphysik, Garching bei München 1966. This report contains 20 figures with the calculated values according to Table 1. Only 4 of these figures are reproduced in the present paper: H I, He II, Ne I, and A I. Reports are available on request from the author.

SRINIVASAN and REES[14] induced me to give more weight to the measurements of RAPP and ENGLANDER-GOLDEN[15]; accordingly I lowered the cross-section for argon (Fig. 4) by approximately 10% up to 100 eV and less than 10% for higher electron energies. There are other papers by SCHRAM, MOUSTAFA, SCHUTTEN, and DE HEER[16] and by ADAMCZYK, BOERBOOM, SCHRAM, and KISTEMAKER[17], who have made measurements

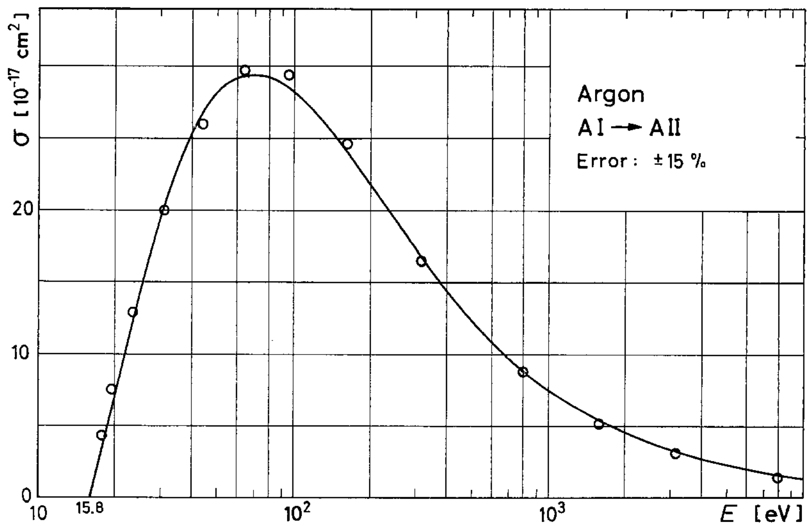

Fig. 4. The curve gives mean experimental values of the electron-impact ionization cross-section for single ionization from the ground state of argon (LOTZ[10], SRINIVASAN *et al.*[14]). For explanations see Fig. 3!

on noble gases. It seems to me that their absolute values might be subject to comparatively high possible errors. I think a correction to the experimental curves of He I, Ne I, Kr I, and Xe I, given in Reference [10], is not yet justified.

Calculated values using the constants of Table 1 are marked by small circles in Figs. 1–4. Values calculated by RUDGE and SCHWARTZ[9] for hydrogen and singly ionized helium are marked by triangles; these values can be reproduced within 1% with the constants of Table 2.

The error given in Figs. 1–4 is a typical value near maximum cross-section (two times probable error). Almost all experimental curves can be

[14] SRINIVASAN, V., and J. A. REES: Brit. J. Appl. Phys. **18**, 59 (1967).

[15] RAPP, D., and P. ENGLANDER-GOLDEN: J. Chem. Phys. **43**, 1464 (1965).

[16] SCHRAM, B. L., H. F. MOUSTAFA, J. SCHUTTEN, and F. J. DE HEER: Physica **32**, 734 (1966).

[17] ADAMCZYK, B., A. J. H. BOERBOOM, B. L. SCHRAM, and J. KISTEMAKER: J. Chem. Phys. **44**, 4640 (1966).

approximated within 10% over the whole energy range up to 10 keV. Exceptions are sodium, rubidium, and cesium, but here the experimental error is large and the calculated values are still within the estimated error limits.

Table 2. *Constants a, b, and c of formula (3) for the electron-impact ionization cross-sections from the ground state of hydrogenlike species given by* RUDGE *and* SCHWARTZ[9]. *The values given by* RUDGE *and* SCHWARTZ *agree with formula (3) within 1% and are marked in Figs. 1 and 2 by triangles. Constant a is given in* 10^{-14} cm^2 $(eV)^2$

Species	Configu-ration	Constants		
		a	b	c
H I	1 *s*	4.27	0.54	1.0
He II	1 *s*	4.4	0.27	0.50
Z=128	1 *s*	4.5	0.04	0.6

Recently GLUPE and MEHLHORN[18] measured ionization cross-sections for electrons of inner shells. These cross-sections can be described by Formula (3) with the constant "a" approximately equal to 4.5×10^{-14} cm^2 (eV)^2. Fine structures of the ionization cross-section curve observed e.g. by FOX[19] and seemingly present with cesium[10,13] cannot be reproduced by the formula proposed.

Note added in proof. D. F. DANCE, M. F. A. HARRISON, and R. D. RUNDEL [Proc. Roy. Soc. A **299**, 525 (1967)] have measured the cross-section for detachment of electrons from H^- by electron impact. Their results can be approximated within 10% in the energy range from 9 to 500 eV by formula (3) with the following constants: $a=1.35\times 10^{-14}$ cm^2 (eV)^2, $b=1.0$, $c=0.055$, $P=0.75$ eV, $q=2$

18 GLUPE, G., and W. MEHLHORN: Verhandl. DPG **1**, 315 (1966).
19 FOX, R. E.: J. Chem. Phys. **35**, 1379 (1961).